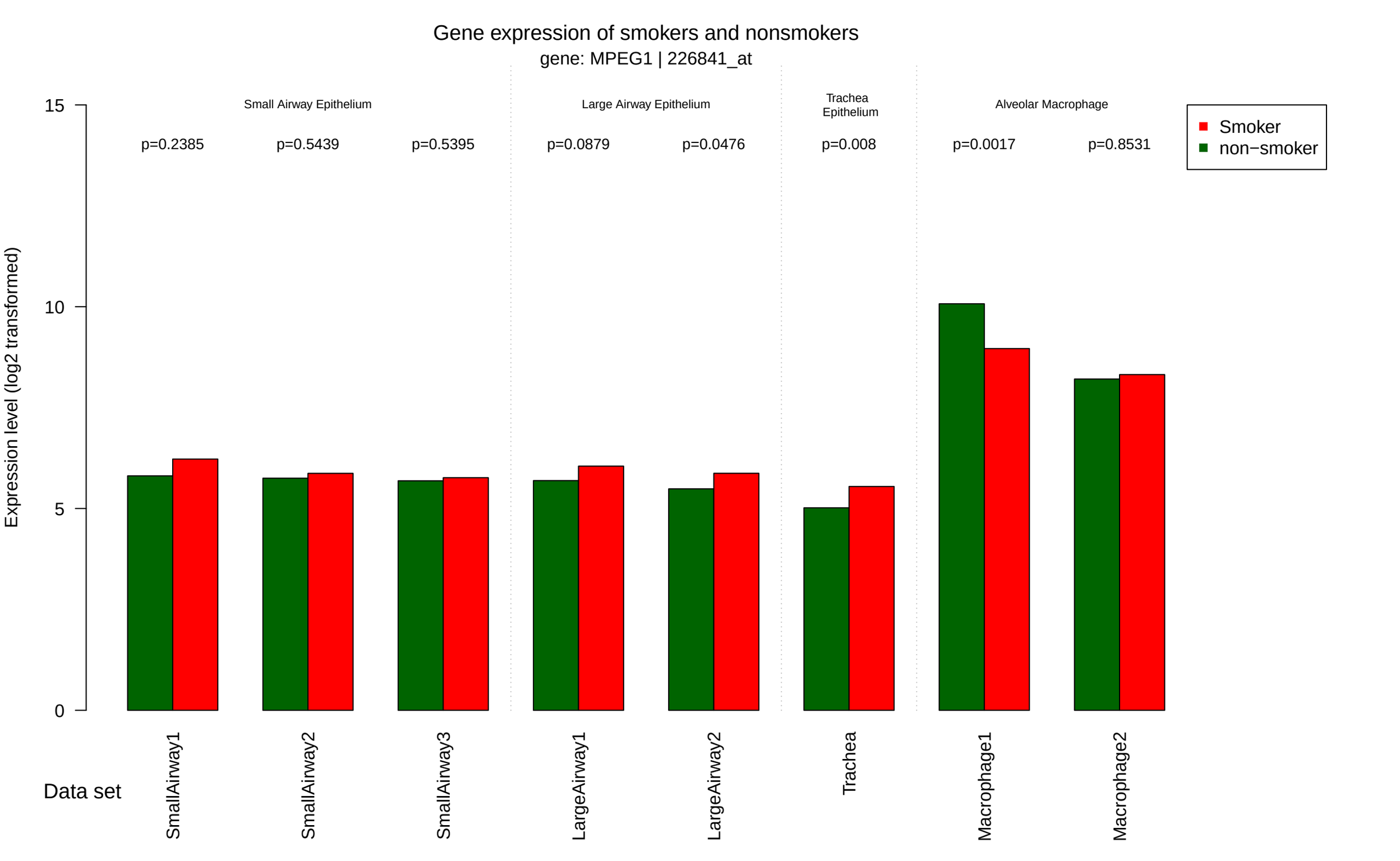
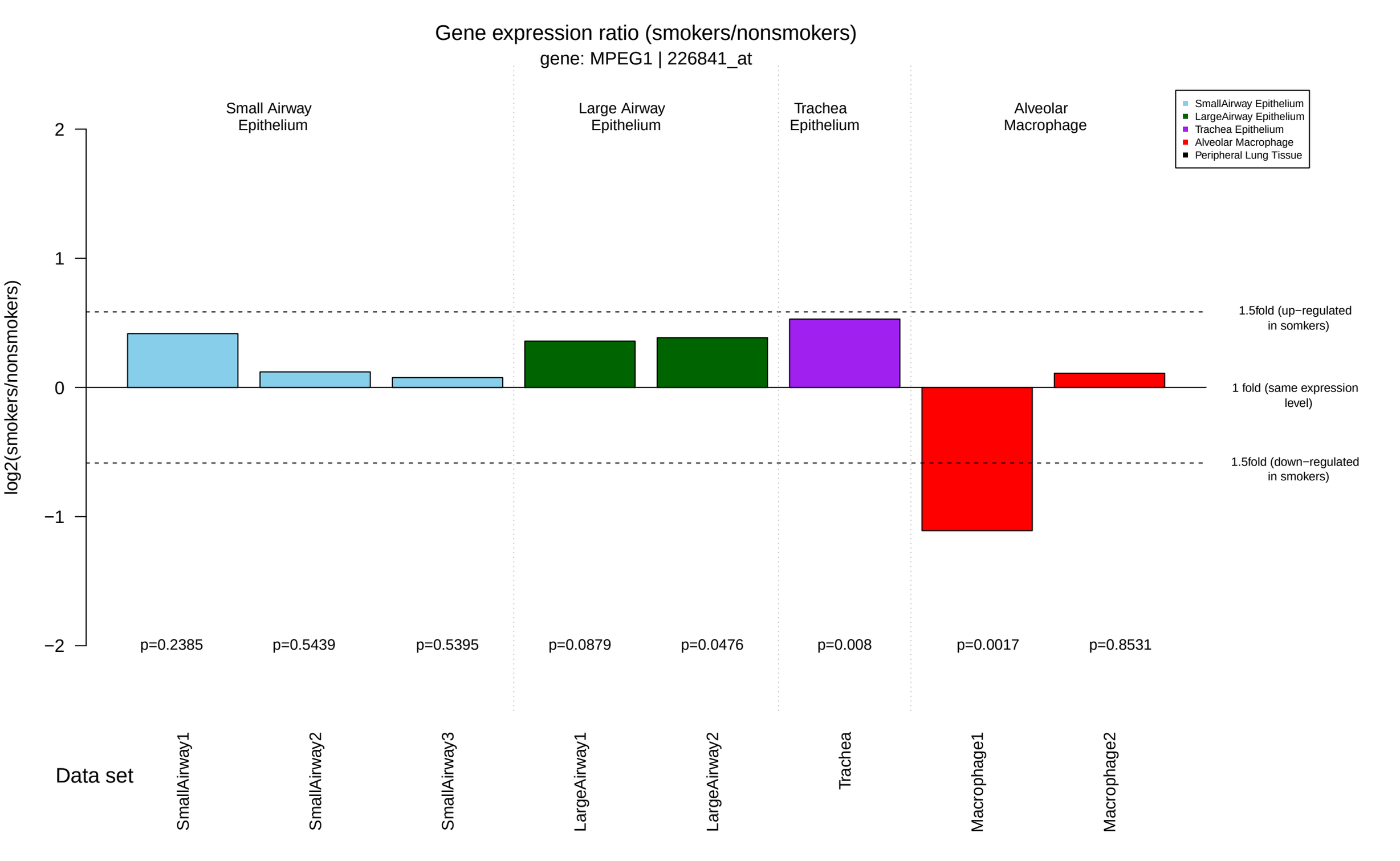
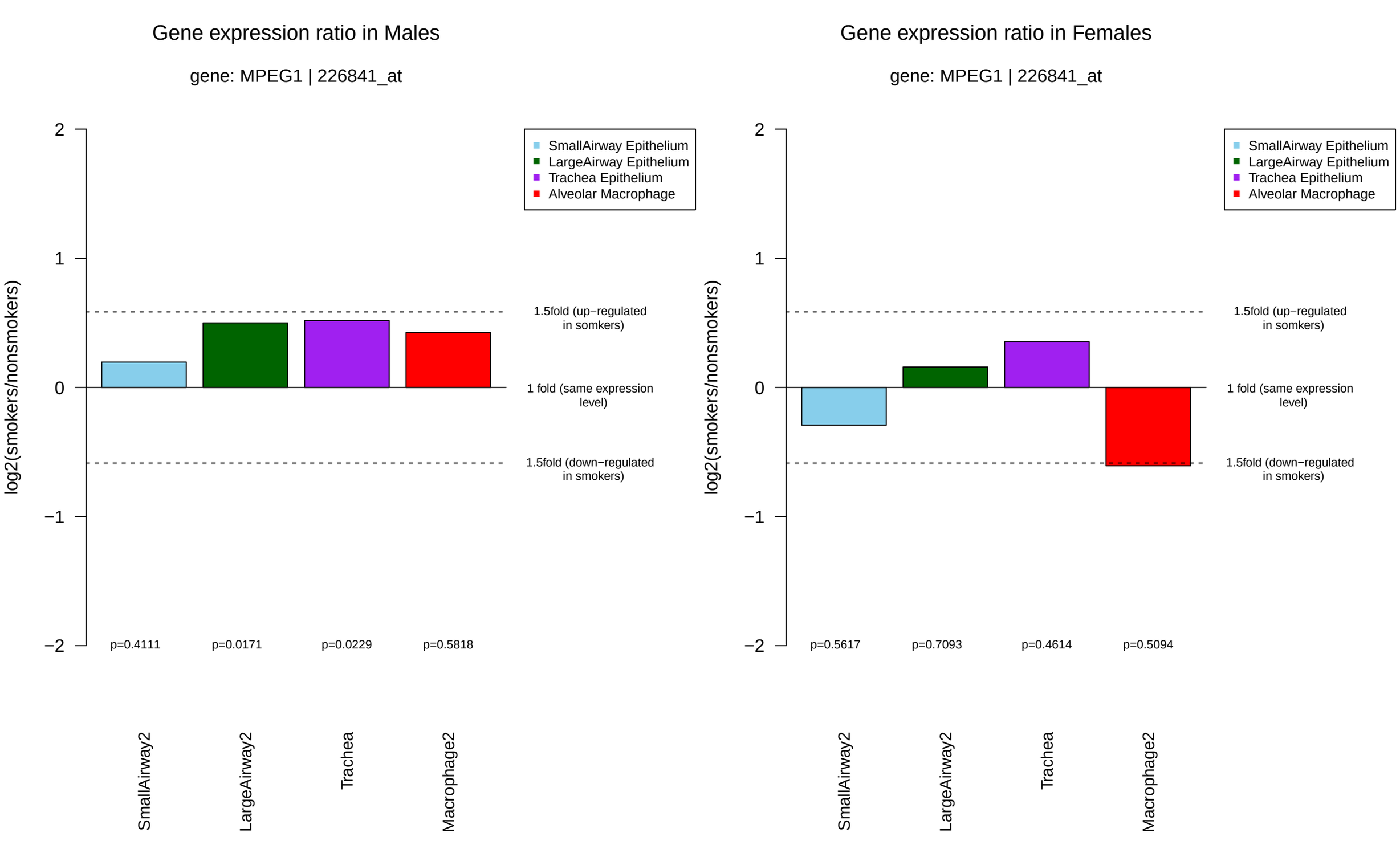
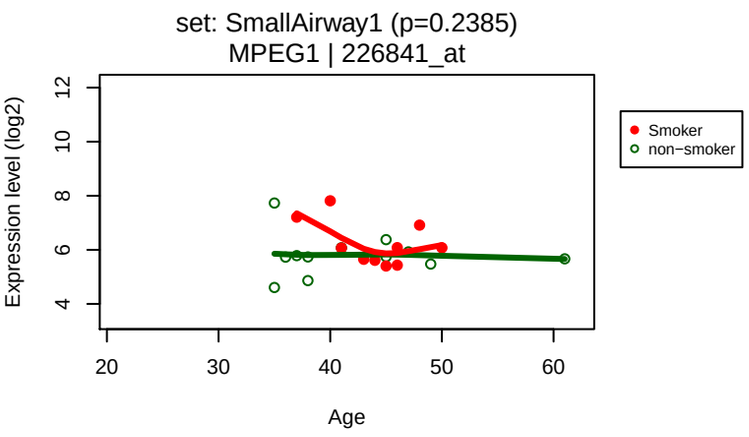
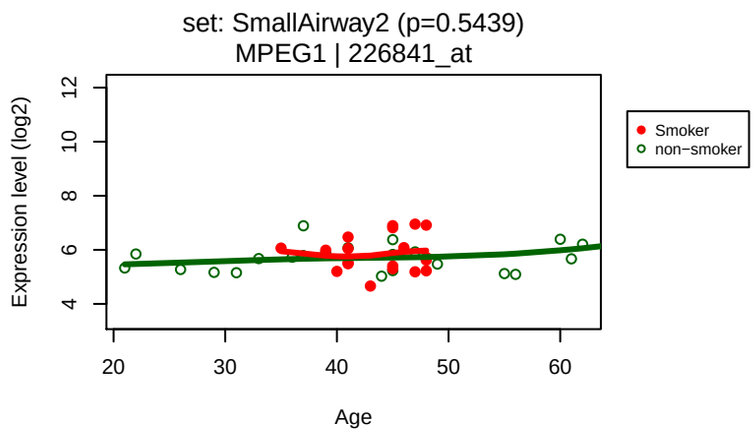
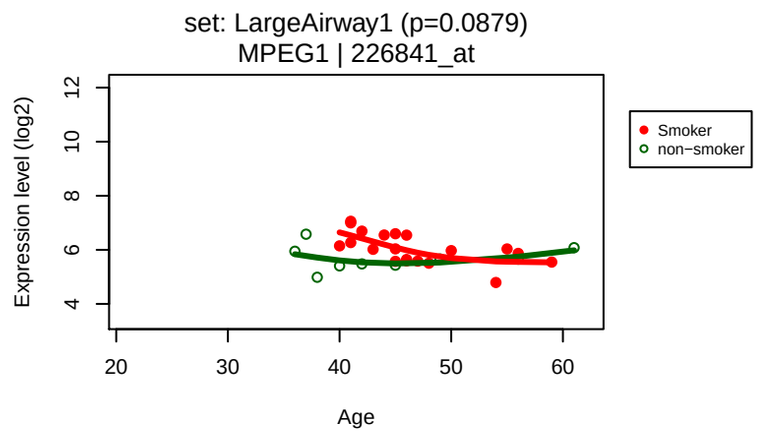

Gene expression of smokers and nonsmokers
gene: MPEG1 | 226841_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2385
p=0.5439
p=0.5395
p=0.0879
p=0.0476
p=0.008
p=0.0017
p=0.8531
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

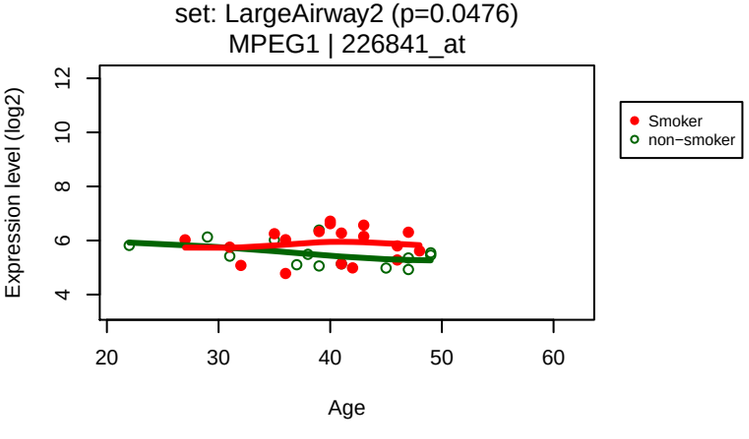

Gene expression ratio (smokers/nonsmokers)
gene: MPEG1 | 226841_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2385
p=0.5439
p=0.5395
p=0.0879
p=0.0476
p=0.008
p=0.0017
p=0.8531
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

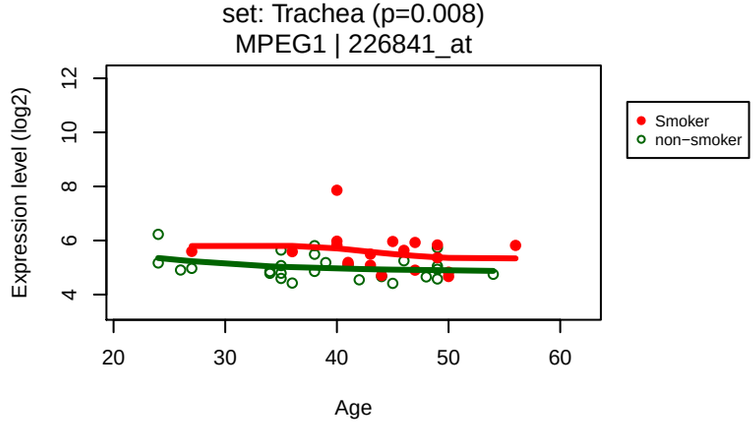

Gene expression ratio in Males
gene: MPEG1 | 226841_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4111
p=0.0171
p=0.0229
p=0.5818
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MPEG1 | 226841_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5617
p=0.7093
p=0.4614
p=0.5094
SmallAirway2
LargeAirway2
Trachea
Macrophage2

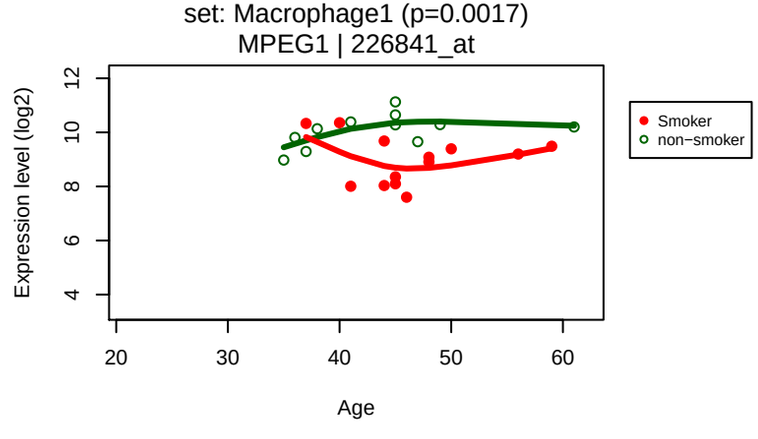

set: SmallAirway1 (p=0.2385)
MPEG1 | 226841_at
Expression level (log2)
Age
Smoker
non−smoker

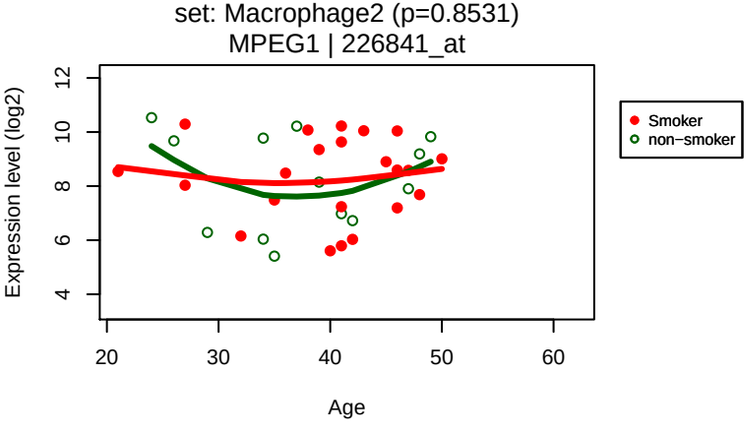

set: SmallAirway2 (p=0.5439)
MPEG1 | 226841_at
Expression level (log2)
Age
Smoker
non−smoker

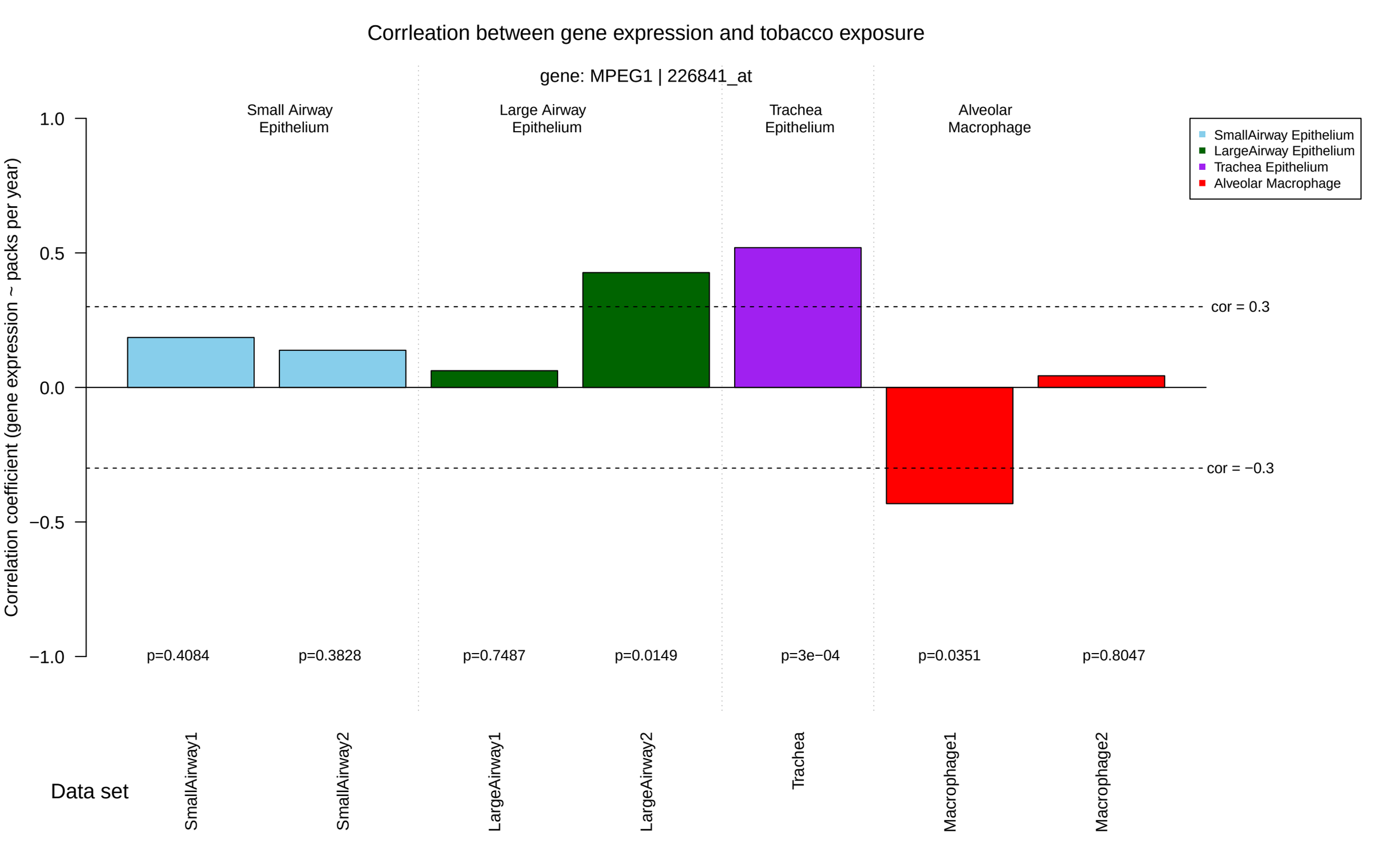

set: LargeAirway1 (p=0.0879)
MPEG1 | 226841_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0476)
MPEG1 | 226841_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.008)
MPEG1 | 226841_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0017)
MPEG1 | 226841_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8531)
MPEG1 | 226841_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MPEG1 | 226841_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4084
p=0.3828
p=0.7487
p=0.0149
p=3e−04
p=0.0351
p=0.8047
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2